WILLIAM FREE HALL

Senior Solutions Architect | Enterprise Cloud & Systems Engineer

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PROFESSIONAL SUMMARY

Results-driven Senior Solutions Architect and Systems Engineer combining 23 years of U.S. Military Special Operations leadership with proven enterprise execution across AWS, Azure, and GCP. Technical SME with demonstrated success leading architecture design sessions, executive briefings, and secure multi-cloud modernization. Deep expertise in zero-trust governance, zero-secret OpenID Connect (OIDC) identity federation, automated CI/CD compliance gates, infrastructure as code (Terraform, Bicep), and high-throughput PySpark Medallion lakehouses. Proven ability to bridge low-level systems engineering with boardroom business outcomes in mission-critical defense, energy, and healthcare ecosystems.

CORE COMPETENCIES & TECHNICAL STACK

- **Architecture & Customer Engagement:** Architecture Design Sessions (ADS), Executive Technical Briefings, Proof-of-Concept (POC) Delivery, Cross-Functional Leadership, Defense & Federal Compliance Alignment.
- **Multi-Cloud & Hybrid:** AWS (Lambda, EC2, S3, IAM, CloudFormation), Microsoft Azure (AKS, Azure Entra ID, App Services, Virtual Networks), GCP (GKE, GCS, Cloud Functions), Multi-Tenant Isolation.
- **IaC & DevSecOps:** Terraform, Bicep, Ansible, GitHub Actions CI/CD, Zero-Secret OIDC Workload Federation, Policy-as-Code, Zero-Downtime Blue/Green Deployments.
- **Distributed Systems & Data:** Databricks, Apache Spark / PySpark (Medallion Architecture), Delta Lake, Unity Catalog Governance, Industrial IoT Telemetry Ingress, Deterministic Event Engines (Append-Only WAL).
- **Containers & Orchestration:** Kubernetes (AKS, GKE), Docker Multi-Stage Builds, Helm Charts, Microservices Architecture.
- **Security & Governance:** Zero-Trust Network Access (ZTNA), Least-Privilege IAM Boundaries, RBAC, SCIF Operations Security, Technical Threat Analysis, Palantir, i2 Analyst's Notebook.

PROFESSIONAL EXPERIENCE

Principal Cloud Solutions Architect & Founder — 2025 – Present

Oaken Sky Cloud Solutions | Remote

- Deliver end-to-end cloud strategy, multi-cloud infrastructure modernization, and enterprise architecture advisory for industrial manufacturing, healthcare, and federal clients.
- Architected automated zero-secret GitHub Actions CI/CD deployment pipelines using Azure Entra ID OpenID Connect (OIDC) Workload Identity Federation, eliminating hardcoded credentials across multi-region deployments.
- Engineered an end-to-end PySpark Medallion lakehouse processing industrial IoT edge telemetry; implemented schema enforcement and automated validation gates matching strict operational availability benchmarks.
- Designed and published CloudWire, an open-source topology analysis engine mapping cross-layer infrastructure dependencies across Terraform, Docker, and Kubernetes configurations.
- Engineered a deterministic, memory-efficient systems event engine with append-only write-ahead log (WAL) persistence, sub-millisecond recovery, and isolated ingress deduplication for distributed microservices.

Technical Lead & Solutions Architect — 2024 – 2025

For Your Service (Partnered with 7 Eagle Group) | Remote / GitHub Enterprise

- Served as principal technical SME driving partner discovery sessions, technical roadmap reviews, and architecture demonstrations translating complex system blueprints for non-technical stakeholders.
- Architected an enterprise PySpark Medallion data pipeline on Databricks utilizing Unity Catalog for role-based governance, secret management, and fine-grained access control across heterogeneous datasets.
- Built serverless and containerized microservice backends on GCP (GKE) and AWS, establishing horizontal autoscaling parameters supporting variable-throughput analytical workloads.

- Implemented modular infrastructure tiers using Terraform, enabling immediate zero-cost staging deployments alongside autoscaling production environments.
- Standardized automated CI/CD release cycles via GitHub Actions, embedding static security analysis, linting gates, and automated infrastructure validation to eliminate downtime.

Cloud Engineer & DevSecOps Analyst **2022 – 2024**

ConocoPhillips | Houston, TX / Remote

- Engineered and maintained multi-tenant cloud environments across AWS, Azure, and GCP, establishing network perimeter security, centralized IAM guardrails, and tenant resource segregation.
- Standardized Infrastructure-as-Code workflows using modular Terraform and Ansible playbooks, cutting provisioning latency and eliminating configuration drift across development, staging, and production.
- Streamlined large-scale analytical data ingestion pipelines, enforcing encryption at rest and in transit, lifecycle data tiering, and high-availability operational standards.
- Deployed real-time system observability architectures with Prometheus, Grafana, and ELK Stack, defining automated alerting thresholds and operational health monitors.

Master Sergeant / Team Sergeant (Duty as Assault Force Sergeant Major) **1994 – 2017**

U.S. Army Special Operations Command (Green Beret) & USMC
Recon | Global Deployments

- Served as Team Sergeant and Master Sergeant, performing duty in the position of Sergeant Major for an operational assault force.
- Planned, resourced, and directed operational and technical intelligence activities for elite tactical teams across multiple high-threat overseas operational rotations.
- Briefed flag-level military officers, interagency executives, and embassy leadership on technical risk assessments, mission-critical infrastructure status, and complex operational intelligence.
- Directed tactical communications, local data networks, and secure compartmented enterprise systems (SCIF); enforced Public Key Infrastructure (PKI), multi-factor token authentication, and role-based access restrictions.
- Exploited enterprise link-analysis platforms (Palantir, i2 Analyst's Notebook) to fuse high-volume, disparate data streams, identifying critical operational nodes and network dependencies.
- Led, mentored, and developed cross-functional elite teams through high-risk operations, cultivating technical agility, strict contingency execution, and mission-first delivery.

EDUCATION & CREDENTIALS

- Bachelor of Science in Cybersecurity | American Military University
- Associate of Arts in Computer Programming Specialist | Northwest Florida State College
- U.S. Army Special Forces Qualification Course (SFQC) - Intelligence Sergeant (18F) | USASOC
- AWS Certified Cloud Practitioner | Amazon Web Services
- Microsoft Certified: Azure Fundamentals (AZ-900) | Microsoft
- U.S. Department of Labor Apprenticeship: Cloud Engineering
- AWS & Azure DevOps Accelerators (8 Months Intensive) | SkillStorm
- Syracuse University Onward to Opportunity (O2O) Fellow | Syracuse University